

Title : Hotel Reservation System

Database Description

The Hotel Reservation System is used to manage hotel operations such as customer bookings, room allocation, payments, and reservation details. The database stores information about guests, rooms, reservations, and payments, helping the hotel efficiently handle room bookings and customer records

Entities, Attributes, and Primary Keys

Entity	Attributes	Primary Key
Guest	Guest_ID, Guest_Name, Gender, Phone, Email, Address	Guest_ID
Room	Room_ID, Room_Number, Room_Type, Price, Status	Room_ID
Reservation	Reservation_ID, Check_In_Date, Check_Out_Date, Number_of_Guests, Guest_ID, Room_ID	Reservation_ID
Payment	Payment_ID, Amount, Payment_Date, Payment_Method, Reservation_ID	Payment_ID

Entity Details

Guest

- PK: Guest_ID
- Attributes: Guest_Name, Gender, Phone, Email, Address

Room

- PK: Room_ID
- Attributes: Room_Number, Room_Type, Price, Status

Reservation

- PK: Reservation_ID
- FK: Guest_ID → Guest(Guest_ID)
- FK: Room_ID → Room(Room_ID)
- Attributes: Check_In_Date, Check_Out_Date, Number_of_Guests

Payment

- PK: Payment_ID
- FK: Reservation_ID → Reservation(Reservation_ID)
- Attributes: Amount, Payment_Date, Payment_Method
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Relationships

1. Guest → Reservation : One Guest can make many Reservations (1:M)
2. Room → Reservation : One Room can have many Reservations over time (1:M)
3. Reservation → Payment : One Reservation has one Payment (1:1)
4. Guest ↔ Room : Many-to-Many relationship through Reservation

Foreign Key

- Guest_ID in Reservation references Guest(Guest_ID)
- Room_ID in Reservation references Room(Room_ID)
- Reservation_ID in Payment references Reservation(Reservation_ID)

HOTEL RESERVATION SYSTEM - ER DIAGRAM

